CRISPR-CAS9 GENE EDITING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on CRISPR-Cas9 Gene Editing

Question 1

10.0 points

Refer to the diagram below that illustrates the Cas9 protein–sgRNA complex docking onto target DNA. Describe the molecular interactions between the sgRNA and the target DNA that are critical for specificity. Identify any key residues or structures in the Cas9 protein involved.

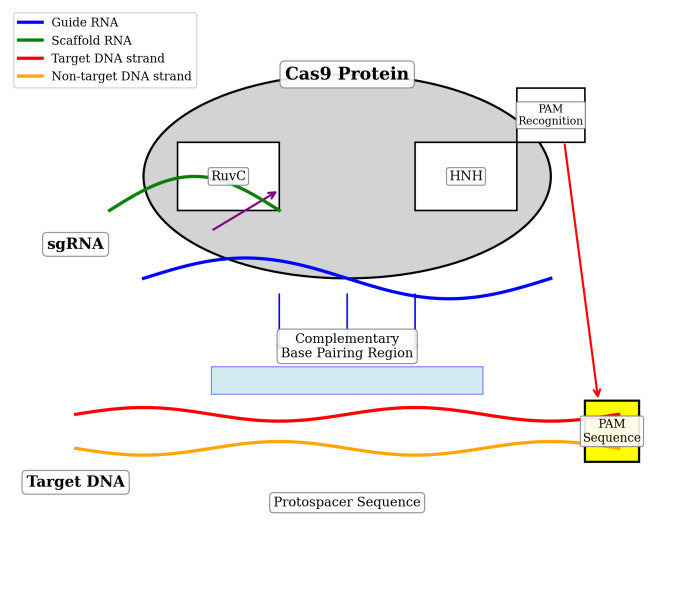

Fig. 1. Diagram for Question 1

Question 2

10.0 points

Examine the diagram below illustrating the PAM sequence recognition and DNA unwinding mechanism in CRISPR-Cas9. Discuss the role of the PAM sequence in Cas9 activation and how DNA unwinding is achieved.

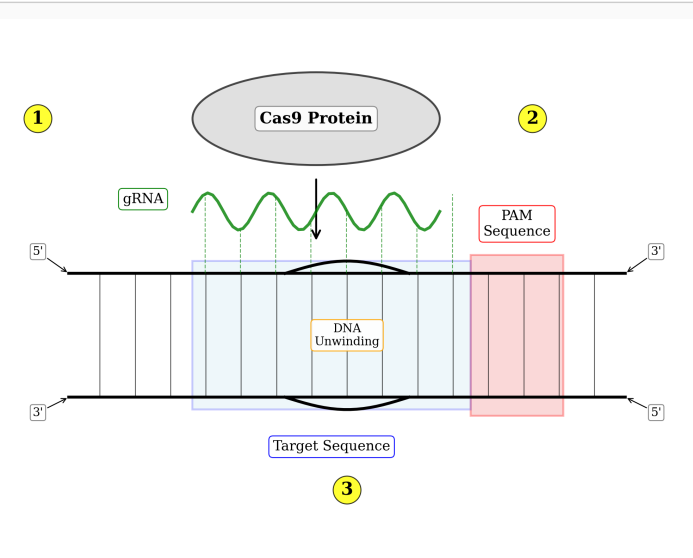

Fig. 2. Diagram for Question 2

Question 3

10.0 points

Examine the diagram below that illustrates DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). Compare these pathways in terms of accuracy and potential applications in gene editing.

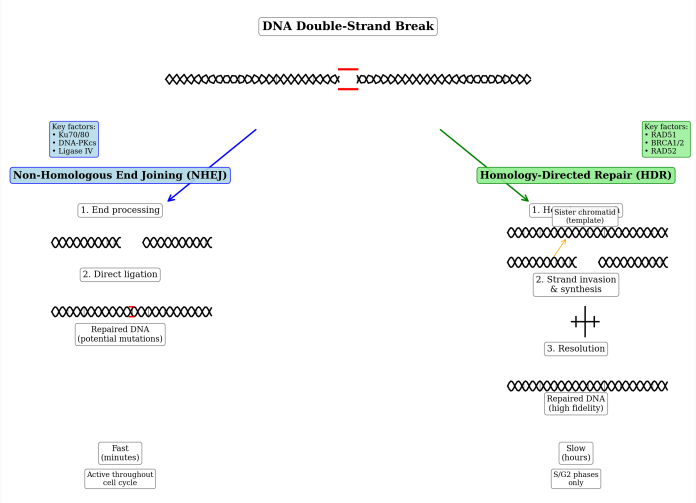

Fig. 3. Diagram for Question 3

Question 4

10.0 points

Examine the mismatch tolerance illustrated in the diagram below for predicting off-target effects in CRISPR-Cas9 gene editing. How do different types of mismatches affect the likelihood of off-target effects?

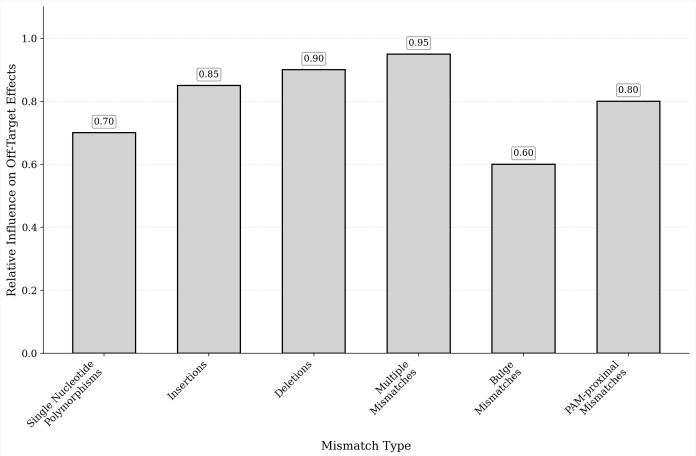

Fig. 4. Diagram for Question 4

Question 5

10.0 points

Analyze the base editing window for cytosine base editors (CBEs) in the diagram below. Describe the CBE's base editing window and discuss the implications for specificity and efficiency in gene editing projects.

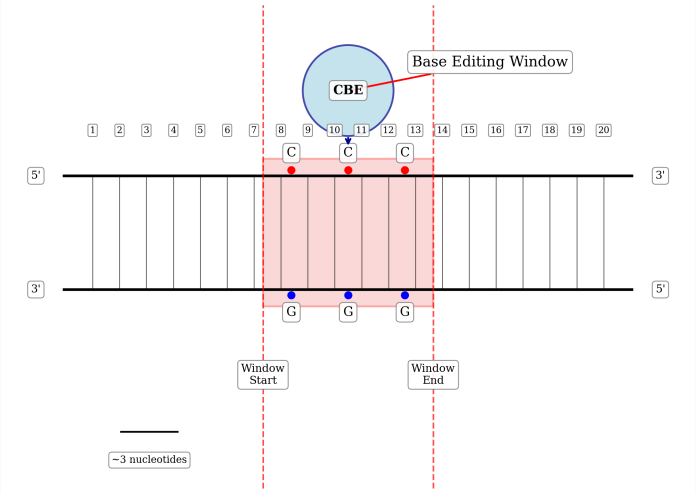

Fig. 5. Diagram for Question 5